Mini-Games Design Brief

How our in-app games work — mechanics, logic, and craving science

For: Product & Design Team | Version 1.0

Why These Games Exist

A craving peaks in 3–5 minutes. It does not disappear because the user wills it away — it fades when the brain is occupied. Our mini-games are not entertainment features. They are precision-designed distractions that occupy the exact mental and physical channels a craving hijacks — the hands, the eyes, and attention.

> **The science:** Kemps & Tiggemann (Flinders University, 2015) found that visuospatial tasks — games that use your eyes and working memory together — reduce craving intensity by an average of 24%. All three of our games work on this principle.

Each game is designed to be fast to start, short to play, and impossible to do mindlessly. That is the design target.

Game 1: Echo Tap — Single Player

What It Is

A rhythm replication game. The game plays a tap sequence. The player replays it back. Think of it like the old Simon Says toy — lights flash in order, you repeat the order — except here it is pure rhythm and tapping.

What the Player Sees and Does

- A simple screen with a visual pulse (expanding circle, waveform, or glowing dots)
- The game plays a sequence of taps — with audio and/or haptic vibration
- The player replicates the sequence by tapping the screen
- Each correct tap gives a satisfying visual or sound response
- Each wrong tap gives a gentle buzz — noticeable, not harsh
- Sequences get longer and faster as the player progresses

How the Game Tracks Progress

Internally, the game maintains three things:

- The sequence — an ordered list of taps and timings it just played
- Your input — your taps, captured in real time and compared against the list, one by one
- Your streak and level — how many sequences in a row you nailed, which raises difficulty

V1 simplification: Track sequence order only, not exact timing precision. This is far easier to build and still delivers the experience. Timing accuracy can be added in V2.

Why It Works as a Craving Tool

- Listening + replicating a rhythm demands real attention — the brain cannot multitask the craving simultaneously
- Rhythmic tapping has documented calming effects on the nervous system (similar to EFT tapping used in stress therapy)
- Short feedback loops — each round is 10–15 seconds — keep restless users hooked, not waiting
- No cultural friction for Indian audiences: rhythm and music are universal touchpoints

Technical Complexity: MEDIUM — Core single-player is buildable. The only nuanced decision is timing precision (order-only vs. millisecond accuracy). Start with order-only.

Game 2: Memory Game

What It Is

A classic card-matching game. A grid of face-down cards, each with a hidden image. Every card has exactly one matching pair somewhere in the grid. Players flip two cards at a time and try to find all the pairs.

Think of it as a filing cabinet where someone shuffled all the folders. You open two drawers at a time, try to remember where things were, and slowly build a mental map until every pair is found.

Version A: Single Player

What the Player Sees and Does

- A grid of face-down cards — 3x4 (12 cards) for a quick craving session, 4x4 (16 cards) for a longer one
- Tap any card to flip it. Tap a second card. If they match: both stay face-up, player scores a point
- If they do not match: both cards flip back face-down after ~1 second. The player tries to remember

- Game ends when all pairs are found. Score = pairs matched. Optional: time taken

How the Game Tracks Progress

Each card has two properties: its position on the grid, and its content (which image it holds). When two cards are flipped, the game checks if they share the same content. If yes: mark both as matched and remove from play. If no: flip both back. Score = total matched pairs.

Why It Works as a Craving Tool

- Pure working memory game — the act of remembering card positions occupies the same mental space a craving needs
- Self-paced: no timer pressure means it works even when the user is stressed (which is often the trigger)
- Satisfying loop: every successful match delivers a small dopamine hit, building its own momentum

Version B: 2 Player

What Changes

- Same grid. Same rules. Both players take turns flipping two cards
- If you find a match, you score a point AND get another turn immediately
- If you do not match, it becomes the other player's turn
- Each player's score is tracked separately. Whoever has more matches when the board is cleared, wins

The Strategic Layer

In single-player, you are competing against your own memory. In 2-player, you watch your opponent's flips and use what they reveal to find your own matches. Information is shared but the game rewards whoever retains it better — a mechanic similar to poker.

Why Multiplayer Matters for This App

Our research identifies social belonging as a primary driver for our 16–27 year old Indian audience. The chai-sutta break is social — it is not really about the cigarette. A 2-player memory game between two friends who are both trying to quit replaces that social ritual without the smoke. That is the real product value here.

Technical Complexity: MEDIUM–HIGH for 2P. Same-room (local/pass-and-play) is straightforward. Online real-time sync requires infrastructure like Firebase or Supabase. Recommendation: build 1P first, then local 2P, then online 2P in V2.

Game 3: Find Match — 2 Player

What It Is

A competitive visual race. Both players see a shared board of mixed icons, patterns, or images. Somewhere on that board is a matching pair. Both players race to find and tap it first. Think of the card game Spot It! — two people looking at the same cards, racing to call out what they share — except digital and faster.

What Both Players See and Do

- A board of visual elements appears on screen
- Both players search for the matching pair simultaneously
- First player to tap the correct match scores a point
- Incorrect tap = short penalty (cannot tap for 2 seconds) — prevents random spam tapping
- New round begins immediately. First to X points wins
- Full game lasts approximately 2–3 minutes across 5–7 rounds

How the Game Decides Who Wins

Simple timestamp logic: the moment a correct tap is registered, the server records the time. The player whose tap arrived first wins that round. Points accumulate. Final score decides the winner after all rounds are played.

Real-Time vs. Turn-Based: The Key Distinction

This is important for the team to understand before we scope the build.

Turn-based: Like chess by post. Player 1 acts, Player 2 receives it later and responds. Does not require both players to be online at the same time. Technically simpler. Works with poor network.

Real-time: Like a phone call. Both players must be live simultaneously. Any lag breaks the fairness of the race. Requires a synchronization server. Technically heavier.

Find Match, as a race game, ideally needs real-time. But there is a practical middle ground:

Recommended V1 approach: Async round model — both players get the same board, play their round on their own device within a 30-second window, and scores are compared at the end of each round. No live sync required. This is how 8 Ball Pool's async mode works — it feels competitive without needing real-time infrastructure.

Why It Works as a Craving Tool

- Visual scanning and pattern matching monopolize attention — the brain cannot crave and race simultaneously
- Competitive pressure creates a different, more productive form of tension that overrides craving anxiety
- Short rounds (10–15 seconds each) fit naturally within the 3–5 minute craving window
- 'Challenge a friend' interaction matches how our audience already uses apps like Ludo King and MPL — zero friction to adopt

Technical Complexity: HIGH — The biggest risk is fairness perception. If one player's board loads slower or their tap registers late, the game feels broken. Async round model eliminates this risk in V1.

At a Glance: All Three Games

Game	Core Mechanic	Cognitive Load	Session Length	Craving Fit	Social	Build Complexity
Echo Tap	Rhythm replication	Moderate	2–4 min	Excellent	None	Medium
Memory (1P)	Visual memory	Low–Medium	3–5 min	Excellent	None	Low
Memory (2P)	Competitive memory	Medium	4–6 min	Good	High	Medium–High
Find Match	Visual race	Medium–High	2–3 min	Excellent	High	High

Directions to Explore

These are not finished features. They are design directions worth discussing as we move into build.

1. Craving-Intensity Adaptive Difficulty

If a user logs a high-intensity craving before opening a game, the game auto-starts at a harder difficulty. The hypothesis: harder cognitive challenge = more effective craving interruption. This is a product layer on top of the game — no changes to game code needed, just a difficulty parameter passed in at launch.

2. Thematic Card Content in Memory Game

Instead of generic icons, cards could show things like '₹100 saved', '24 hours clean', or 'deep breath'. The images the brain processes during play do register. This keeps the quit context alive without being preachy.

3. Post-Game Micro-Reflection

One question after every game: 'Did the craving pass?' Yes / No. Over time this builds the user's own evidence that the games work for them — which is more motivating than any push notification we could send. It also gives us data.

4. Game Streaks as a Parallel Progress Metric

For users who slip on the quit-day streak, game streaks keep them engaged. '5 cravings fought with a game this week' is a tangible, positive metric even if they are still working on quitting. Separating the two gives users more than one way to feel like they are winning.

5. Build Sequence Recommendation

Suggested phasing based on complexity and impact:

- Phase 1 — Memory Game (Single Player): lowest complexity, proven craving tool, can ship fast
- Phase 2 — Echo Tap (Single Player): medium complexity, unique mechanic, strong rhythm-centric differentiation
- Phase 3 — Memory Game (2 Player, local): adds social without full online infrastructure
- Phase 4 — Find Match (2 Player, async): highest complexity, highest social payoff